

```
import javax.swing.*;

import java.awt.*;

import java.awt.event.*;

import java.io.*;

import java.net.*;

/*
 * Combined Chat Application UI
 * Run once with role = "server"
 * Run again with role = "client"
 */

public class ChatApplication extends JFrame implements ActionListener {

    private JTextArea chatArea;

    private JTextField inputField;

    private JButton sendButton;

    private Socket socket;

    private BufferedReader reader;

    private PrintWriter writer;

    private String role;

    public ChatApplication(String role) {

        this.role = role.toLowerCase();

        // ----- UI Setup -----

        setTitle("Chat Application - " + role.toUpperCase());
```

```
setSize(500, 600);

setDefaultCloseOperation(EXIT_ON_CLOSE);

setLayout(new BorderLayout());

getContentPane().setBackground(Color.WHITE);


chatArea = new JTextArea();

chatArea.setEditable(false);

chatArea.setFont(new Font("SansSerif", Font.PLAIN, 15));

JScrollPane scrollPane = new JScrollPane(chatArea);

add(scrollPane, BorderLayout.CENTER);


JPanel bottomPanel = new JPanel(new BorderLayout());

inputField = new JTextField();

inputField.setFont(new Font("SansSerif", Font.PLAIN, 15));

sendButton = new JButton("Send");

sendButton.setFont(new Font("SansSerif", Font.BOLD, 14));

sendButton.setBackground(new Color(0, 102, 204));

sendButton.setForeground(Color.WHITE);

sendButton.setFocusPainted(false);


bottomPanel.add(inputField, BorderLayout.CENTER);

bottomPanel.add(sendButton, BorderLayout.EAST);

add(bottomPanel, BorderLayout.SOUTH);


sendButton.addActionListener(this);

inputField.addActionListener(this);
```

```

setVisible(true);

// ----- Connection -----

new Thread(() -> setupConnection()).start();
}

private void setupConnection() {
    try {
        if (role.equals("server")) {
            ServerSocket serverSocket = new ServerSocket(5000);
            chatArea.append("Server started. Waiting for client...\n");
            socket = serverSocket.accept();
            chatArea.append("Client connected!\n");
        } else {
            String serverIP = JOptionPane.showInputDialog(this, "Enter Server IP (127.0.0.1 for local):");
            chatArea.append("Connecting to server...\n");
            socket = new Socket(serverIP, 5000);
            chatArea.append("Connected to server!\n");
        }

        reader = new BufferedReader(new InputStreamReader(socket.getInputStream()));
        writer = new PrintWriter(socket.getOutputStream(), true);

        // Thread to receive messages

```

```

new Thread(() -> {

    String msg;

    try {

        while ((msg = reader.readLine()) != null) {

            chatArea.append(msg + "\n");

        }

    } catch (IOException e) {

        chatArea.append("Connection closed.\n");

    }

}).start();

} catch (Exception e) {

    chatArea.append("Error setting up connection: " + e.getMessage() + "\n");

}

}

@Override

public void actionPerformed(ActionEvent e) {

    String message = inputField.getText().trim();

    if (!message.isEmpty()) {

        chatArea.append("You: " + message + "\n");

        writer.println((role.equals("server") ? "Server" : "Client") + ": " + message);

        inputField.setText("");

    }

}

```

```
public static void main(String[] args) {  
    String[] options = {"Server", "Client"};  
    String role = (String) JOptionPane.showInputDialog(  
        null,  
        "Select your role:",  
        "Chat Application",  
        JOptionPane.QUESTION_MESSAGE,  
        null,  
        options,  
        options[0]  
    );  
  
    if (role != null) {  
        SwingUtilities.invokeLater(() -> new ChatApplication(role));  
    } else {  
        System.exit(0);  
    }  
}
```